SUBJECT : Digital Image Processing

UNIT : 1 DIGITAL IMAGE FUNDAMENTALS

LECTURE : 1

TOPIC : Introduction to Image Processing & Applications

Dr. Shashi Kumar Singh, Director (R&D)

- **OBJECTIVES: To learn digital image fundamentals.**
- **To be exposed to simple image processing techniques.**
- **To be familiar with image compression and segmentation techniques**
- **To acquire knowledge of real time image processing for industrial applications**

Syllabus UNIT - I DIGITAL IMAGE FUNDAMENTALS

(Introduction to Image Processing: Digital Image representation, Sampling & Quantization, Steps in image Processing, Image acquisition, color image representation.)

OUTCOMES:

At the end of the course, the student should be able to:

- ✓ Understand the image enhancement techniques
- ✓ Understand the concept of restoration and segmentation
- ✓ Understand wavelets and image compression

TEXT BOOK:

1. Rafael C. Gonzales, Richard E. Woods, "Digital Image Processing", Third Edition, Pearson Education, 2010.

REFERENCES:

1. Rafael C. Gonzalez, Richard E. Woods, Steven L. Eddins, "Digital Image Processing Using MATLAB", Third Edition Tata Mc Graw Hill Pvt. Ltd., 2011.
2. Anil Jain K. "Fundamentals of Digital Image Processing", PHI Learning Pvt. Ltd., 2011.
3. Willliam K Pratt, "Digital Image Processing", John Willey, 2002.
4. Malay K. Pakhira, "Digital Image Processing and Pattern Recognition", First Edition, PHI Learning Pvt. Ltd., 2011

Pre-requisite: Basic knowledge of Signals & Systems, Digital Signal Processing and Digital Design, Mathematics (transforms)

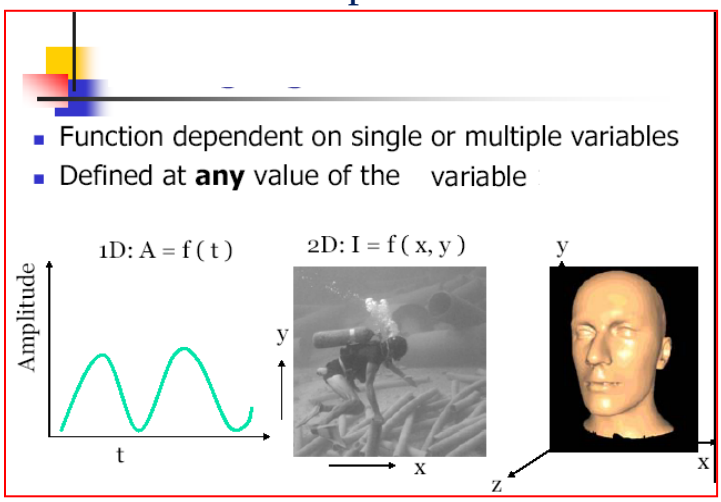
Goals of DIP

- To retrieve the information that is hidden in the image, to analyze the signal.
- To change the signal information suitable to requirement.
- **Noise removal, Image Enhancement, Recognition, Real time image operation, Data Compression, multiresolution (a digital desktop microscope), data transfer, feature extraction**

LEARNING OBJECTIVES in unit1:

This unit provides an overview of the image –processing system which includes various elements like image sampling, quantization, Basic steps in image processing, image formation, storage and display. After completing this unit, the reader is expected to be familiar with the following concepts:

1. Digital Image
2. Image sampling
3. Image sensors
4. Different steps in image processing
5. Image formation



DIGITAL IMAGE FUNDAMENTALS:

Image – A pictorial data which contains information. It is function of intensity on space i.e. (x,y) co ordinate.

I = f (x,y)

2 D signal depends on x ,y.

Diigital Image- A two dimensional pictorial data stored in discrete set of number or digital number is called digital image.

A two dimensional function f (x,y), x & y are spatial co- ordinates.

The amplitude of f is called intensity or gray level at point (x,y). It is function of independent variables such as distances, positions . An image contains information.

- Changing or analyzing information stored in the form of pixels that is measured as discrete sequences of numbers is called **"*Digital Image processing*".**
- **Image-2D Signals: A 2D *signal* is a description of how one parameter is related to another parameter containing information. Intensity function depends on x,y co ordinates.**

  - It is a function of independent variables such distance, position

  - A *Image* carries information.
- **Examples: image and video.**
- **Digital image – The intensity and pictorial information stored in discrete set of numbers (digital numbers) is digital image .**

Changing or analyzing information stored in the form of pixels that is measured as discrete sequences of numbers is called "*Digital Image processing*".

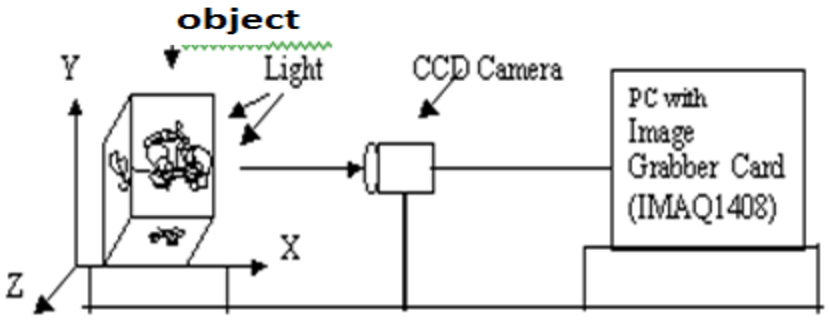
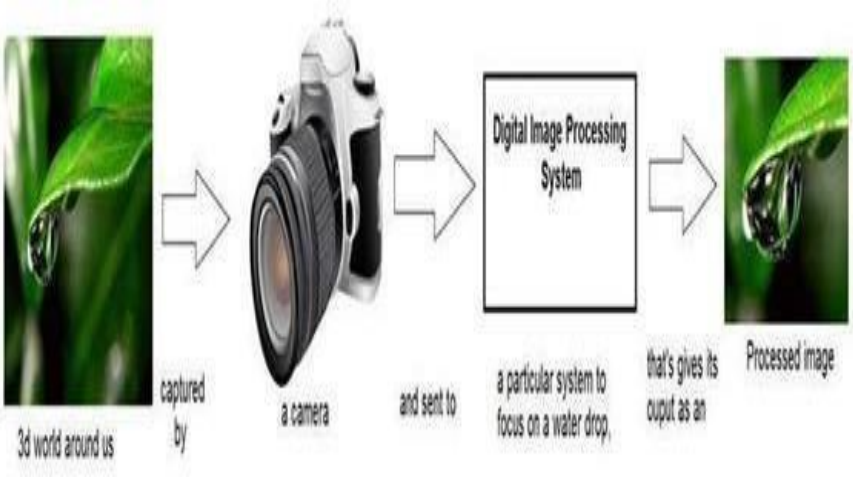


Fig.1 Schematic Diagram of Detection system of contaminants from industrial product

Fig.1 Basic image acquisition and processing set up



The field of digital image processing refers to processing digital images by means of digital computer. Digital image is composed of a finite number of elements, each of which has a particular location and value. These elements are called picture elements, image elements, pels and pixels. Pixel is the term used most widely to denote the elements of digital image.
An image is a two-dimensional function that represents a measure of some characteristic such as brightness or color of a viewed scene. An image is a projection of a 3- D scene into a 2D projection plane.

An image may be defined as a two-dimensional function ***f(x,y),*** where x and y are spatial (plane) coordinates, and the amplitude to ***f*** at any pair of coordinates **(x,y)** is called the intensity of the image at that point. The term **gray level** is used often to refer to the intensity of monochrome images.
Color images are formed by a combination of individual 2-D images. For example, the RGB color system, a color image consists of three (red, green and blue) individual component images. For this reason, many of the techniques developed for monochrome images can be

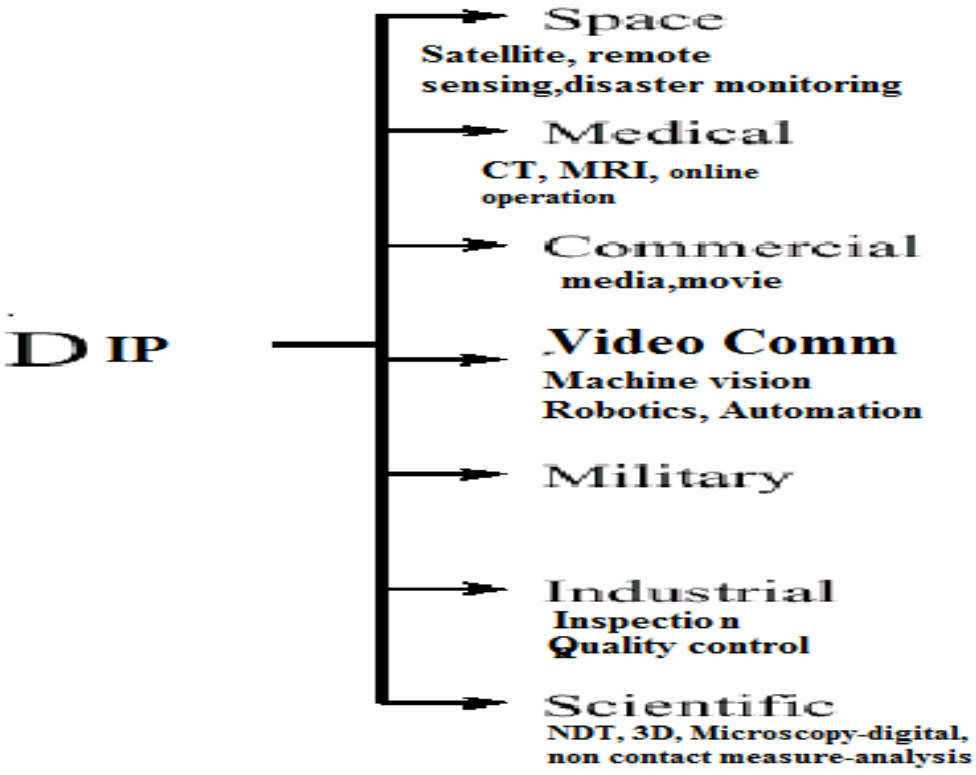
extended to color images by processing the three component images individually.

An image may be continuous with respect to the x- and y-coordinates and also in amplitude. Converting such an image to digital form requires that the coordinates, as well as the amplitude, be digitized.

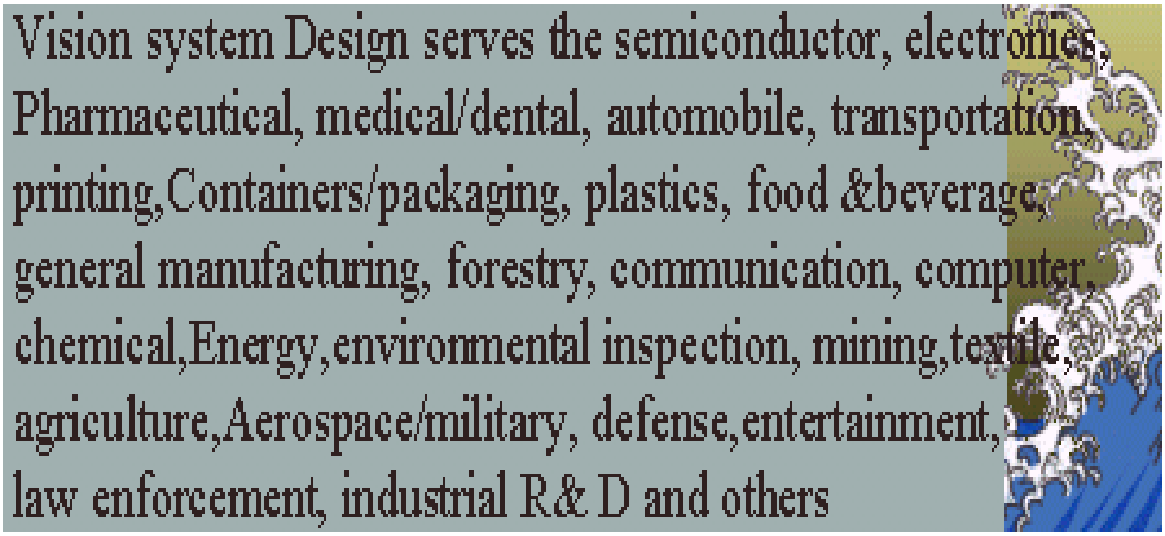
Pixel: In a digital image, all the coordinates on 2-d function and the corresponding values are finite. Each value available in every location is considered as a pixel. In other words, a pixel is the smallest part of an image. So a digital image can be thought as 2-d array of pixels.

Gray Level: Each pixel has some intensity value which is called Gray level or Gray value. These values are usually represented in 8-bit int. So the range of values from 0 to 255. The values near to 0 indicates darker regions and the values near 255 represent brighter regions.

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Key areas



APPLICATIONS OF DIGITAL IMAGE PROCESSING:

Since digital image processing has very wide applications and almost all of the technical fields are impacted by DIP, we will just discuss some of the major applications of DIP.
Digital image processing has a broad spectrum of applications, such as

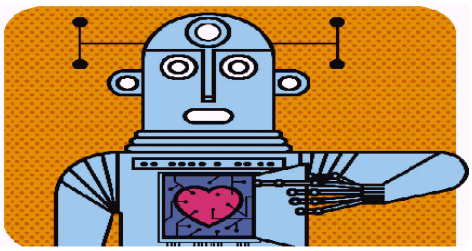
1. Remote sensing via satellites and other spacecrafts
2. Image transmission and storage for business applications
3. Medical processing
4. RADAR (Radio Detection and Ranging)
5. SONAR (Sound Navigation and Ranging)
6. Acoustic Image Processing (The study of underwater sound is known asUnderwater Acoustics or Hydro-Acoustics)
7. Robotics and automated inspection of industrial partsImages acquired by satellites are useful in trackingof objects

1. Earth resources
2. Geographical mapping
3. Prediction of agricultural crops
4. Urban growth and weather monitoring
5. Flood and fire control and many other environmental applicationsSpace image applications include:

1. Recognition and analysis of objects contained in images obtained from deepspace-probe emissions.
2. Image transmission and storage applications occur in broad cast television
3. Teleconferencing
4. Transmission of facsimile images (Printed documents and graphics) foroffice automation
5. Communication over computer networks
6. Closed-circuit television-based security monitoring systems and
7. In military communications

Medical applications:

1. Processing of chest X-rays
2. MRI, CT Scan
3. Projection images of trans axial tomography and
4. Medical images that occur in radiology nuclear magnetic resonance (NMR)
5. Ultrasonics canning





Protein crystal imaging



Image analysis

Machine act like human-

To see, analyze, make the decision about error,

discontinuity of a system and then control or remove

Career and current fields

IMAGE PROCESSING TOOLBOX (IPT):

It is a collection of functions that extend the capability of the MATLAB numeric computing environment. These functions, and the expressiveness of the MATLAB language, make many image-processing operations easy to write in a compact, clear manner, thus providing an ideal software prototyping environment for the solution of image processing problem.

COMPONENTS OF IMAGE PROCESSING SYSTEM:

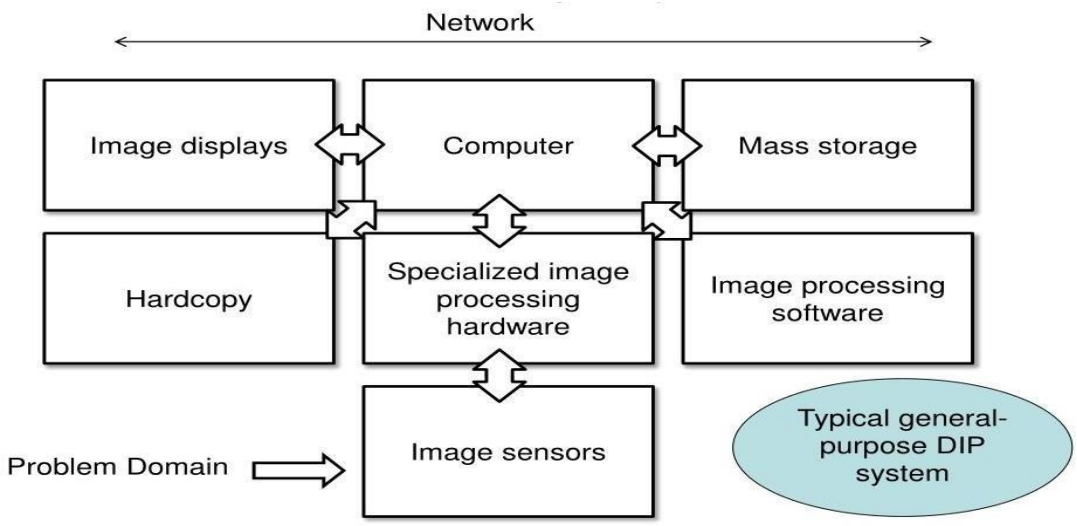


Fig2: Components of Image processing System

Hardware-

Specialize Image Processing Hardware: It consists of the digitizer just mentioned, plus hardware that performs other primitive operations such as an arithmetic logic unit, which performs arithmetic such addition and subtraction and logical operations in parallel on images.

Computer: It is a general-purpose computer and can range from a PC to a supercomputer depending on the application. In dedicated applications, sometimes specially designed computer is used to achieve a required levelof performance

Software: It consists of specialized modules that perform specific tasks a well-designed package also includes capability for the user to write code, as a minimum, utilizes the specialized module. More sophisticated software packages allow the integration of these modules.

Mass Storage: This capability is a must in image processing applications. An image of size 1024 x1024 pixels, in which the intensity of each pixel is an 8- bit quantity requires one Megabytes of storage space if the image is not compressed. Image processing applications falls into three principal categories of storage.

- Short term storage for use duringprocessing
- On line storage for relatively fastretrieval

- Archival storage such as magnetic tapes anddisks

Image Display: Image displays in use today are mainly color TV monitors.These monitors are driven by the outputs of image and graphics displays cards that are an integral part of computer system.

Hardcopy Devices: The devices for recording image include laser printers, film cameras, heat sensitive devices inkjet units and digital units such as optical and CD ROM disk. Films provide the highest possible resolution, but paper is the obvious medium of choice for written applications.

Networking: It is almost a default function in any computer system in usetoday because of the large amount of data inherent in image processing applications. The key consideration in image transmission bandwidth.

Lecture-2
Topic-Digital Image representation-

Representing Digital Images: The result of sampling and quantization is matrix of real numbers. Assume that an image f(x,y) is sampled so that the resulting digital image has M rows and N Columns. The values of the coordinates (x,y) now become discrete quantities thus the value of the coordinates at origin become (x,y) = (0,0) The next Coordinates value along the first signify the image along the first row. it does not mean that these are the actual values of physical coordinates when the image was sampled. Thus, the right side of the matrix represents a digital element, pixel or pel. The matrix can be represented in the following form as well. The sampling process may be viewed as partitioning the XY plane into a grid with the coordinates of the center of each grid being a pair of elements from the Cartesian products $Z2$ which is the set of all ordered pair of elements (Zi, Zj) with Zi and Zj being integers from Z.

$$f(x,y) = \begin{bmatrix} f(0,0) & f(0,1) & f(0,2) & \dots & f(0,N-1) \\ f(1,0) & f(1,1) & f(1,2) & \dots & f(1,N-1) \\ \vdots & \vdots & \vdots & & \vdots \\ f(M-1,0) & f(M-1,1) & f(M-1,2) & \dots & f(M-1,N-1) \end{bmatrix}$$

Hence f(x,y) is a digital image ifgray level (that is, a real number from the set of real number R) to each distinct pair of coordinates (x,y). This functional assignment is the quantization process. If the gray levels are also integers, Z replaces R, the and a digital image become a 2D function

whose coordinates and she amplitude value are integers. Due to processing storage and hardware consideration, the number gray levels typically are an integer power of 2.

$$L=2^k$$

Then, the number 'b' of bites required to store a digital image is

$$b=M$$

N kWhen M=N, the equation become

$$b=N^2*k$$

When an image can have 2^k gray levels, it is referred to as "k- bit". Animage with 256 possible gray levels is called an "8- bit image" ($256=2^8$).

Image is function of intensity on x,y
Position, i.e. pixel.

It is a 2 D function.
Fig. Image I= f (x,y)

Spatial and Gray level Resolution:

Spatial resolution is the smallest discernible details are an image. Suppose a chart can be constructed with vertical lines of width w with the space

between the also having width W, so a line pair consists of one such line and its adjacent space thus. The width of the line pair is 2w and there is 1/2w line pair per unit distance resolution is simply the smallest number of discernible line pair unitdistance.

Gray levels resolution refers to smallest discernible change

in gray levels. Measuring discernible change in gray levels is a highly subjective process reducing the number of bits R while repairing the spatial resolution constant creates the problem of false contouring.It is caused by the use of an insufficient number of gray levels on the smooth areas of the digital image. It is called so because the rides resemble top graphics contours in a map. It is generally quite visible in image displayed using 16 or less uniformly spaced gray levels.

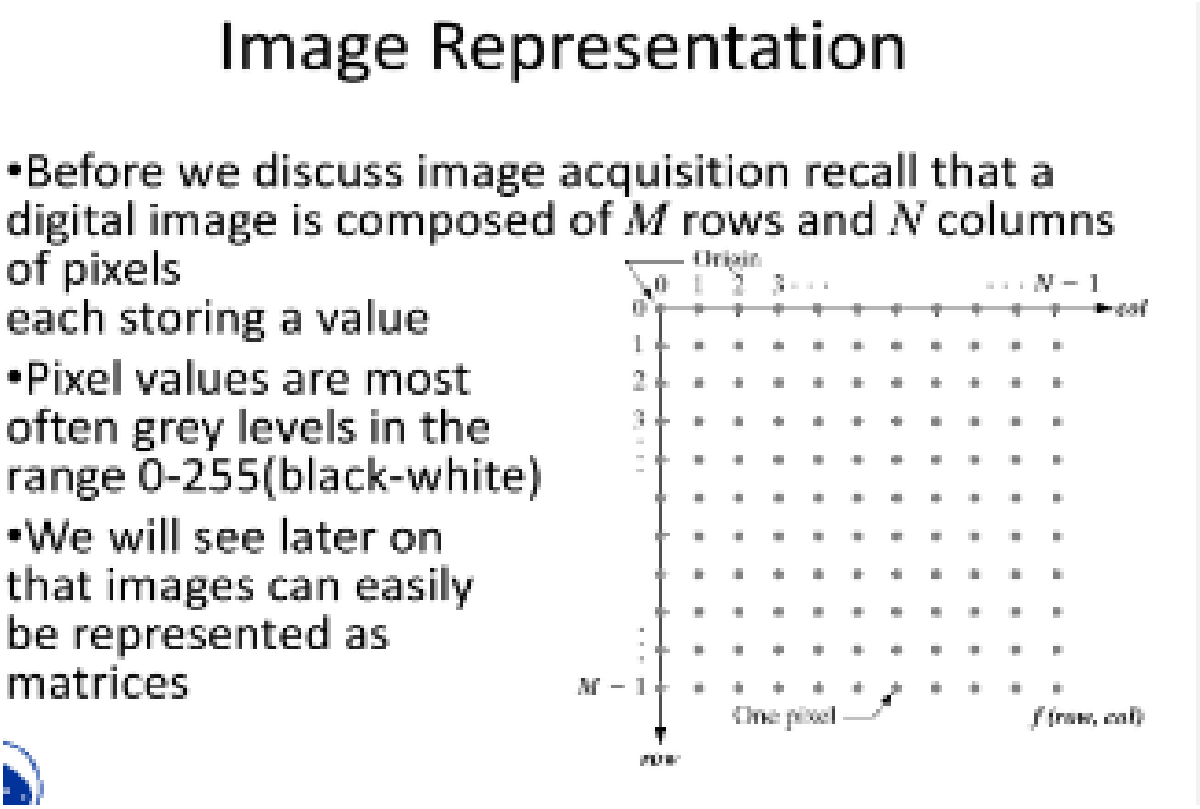


Fig.3a Image representation

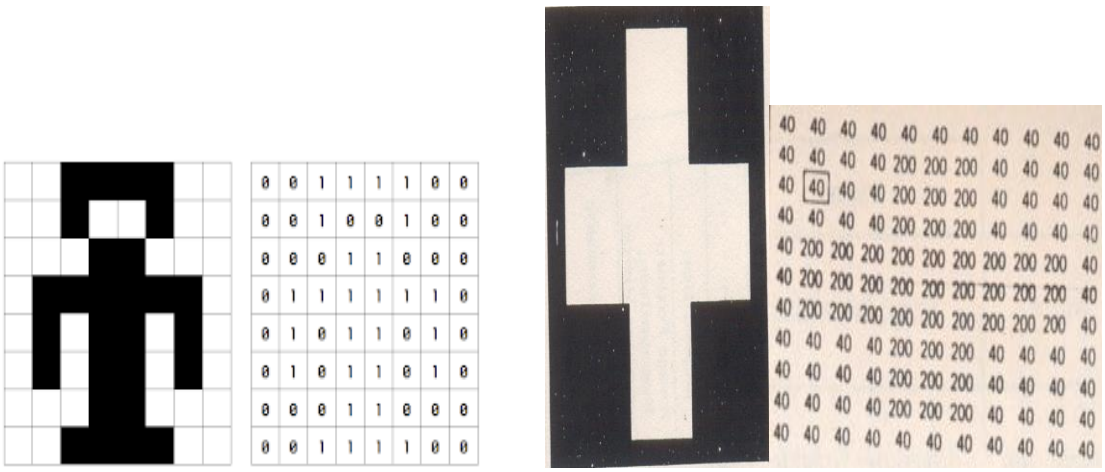


Fig. Demonstration of numerical value assigned to each pixel in above image

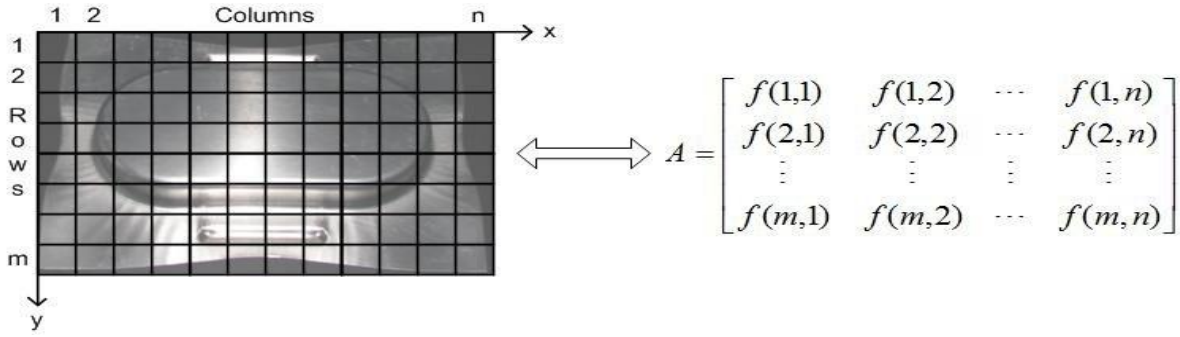


Fig.3b Image representation

$$f(x,y) = \begin{bmatrix} f(0,0) & f(0,1) & f(0,2) & \dots & f(0,N-1) \\ f(1,0) & f(1,1) & f(1,2) & \dots & f(1,N-1) \\ \vdots & \vdots & \vdots & & \vdots \\ f(M-1,0) & f(M-1,1) & f(M-1,2) & \dots & f(M-1,N-1) \end{bmatrix}$$

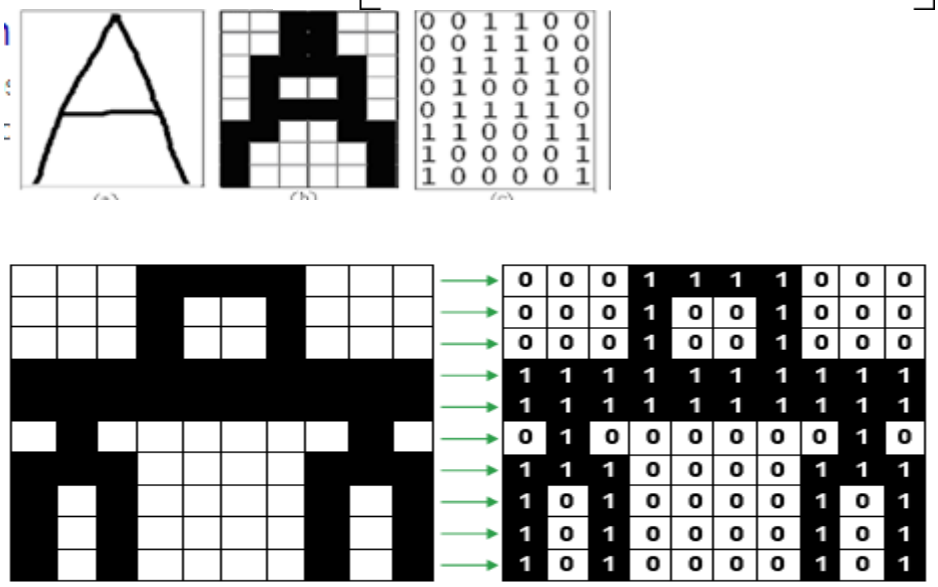


Fig.3c Image representation

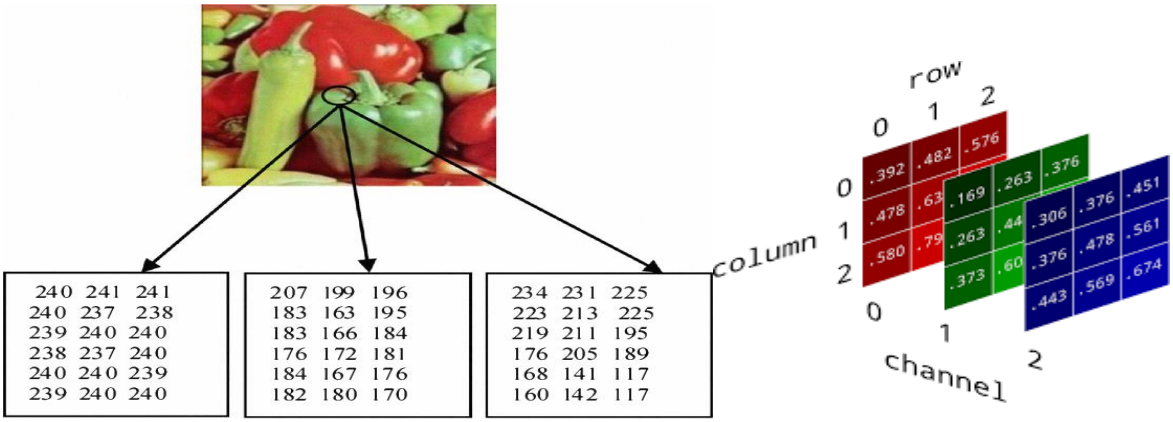


Fig.3d colorImage representation

Digital Images:

We will use two principal ways to represent digital images. Assume that an image f(x, y) is sampled so that the resulting digital image has M rows and N columns. The values of the coordinates (x, y) now become discrete quantities. For notational clarity and convenience, we shall use integer values for these discrete coordinates. Thus, the values of the coordinates at the origin are (x, y) = (0, 0). The next coordinate values along the first row of the image are represented as (x, y) = (0, 1). It is important to keep in mind that the notation (0, 1) is used to signify the second sample along the first row. It does not mean that these are the actual values of physical coordinates when the image was sampled. Figure 4 shows the coordinate convention used.

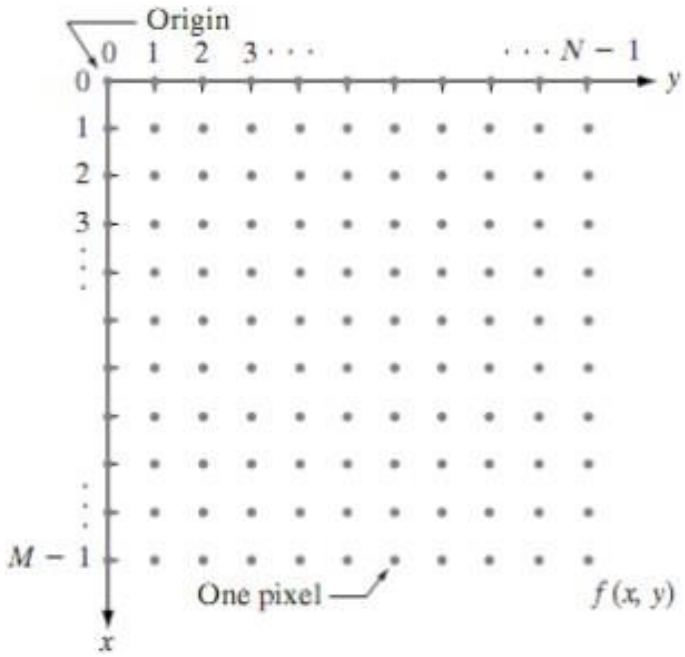


Fig 4 Coordinate convention used to represent digital images

The notation introduced in the preceding paragraph allows us to write the complete M*N digital image in the following compact matrix form:

The right side of this equation is by definition a digital image. Each element of this matrix array is called an image element, picture element, pixel, or pel.

Spatial and Gray level resolution:

Spatial resolution is the smallest discernible details are an image. Suppose a chart can be constructed with vertical lines of width w with the space between the also having width W, so a line pair consists of one such line and its adjacent space thus. The width of the line pair is 2w and there is 1/2w line pair per unit distance resolution is simply the smallest number of discernible line pair unit distance.

Gray levels resolution refers to smallest discernible change in gray levels. Measuring discernible change in gray levels is a highly subjective process reducing the number of bits R while repairing the spatial resolution constant creates the problem of false contouring. It is caused by the use of an insufficient number of gray levels on the smooth areas of the digital image . It is called so

because the rides resemble top graphics contours in a map. It is generally quite visible in image displayed using 16 or less uniformly spaced gray levels.

Relationship between Pixels:

We consider several important relationships between pixels in a digital image.

Neighbors of a Pixel:

A pixel *p* at coordinates (*x*,*y*) has four horizontal and verticalneighbors whose coordinates are givenby:

(x+1,y), (x-1, y), (x, y+1), (x,y-1)

	(x, y-1)	
(x-1, y)	P (x,y)	(x+1, y)
	(x, y+1)	

This set of pixels, called the 4-neighbors or p, is denoted by $N_4(p)$. Each

pixel is one unit distance from (x,y) and some of the neighbors of p lie outside the digital image if (x,y) is on the border of the image. The four diagonal neighbors of p have coordinates and are denoted by $N_D(p)$.

(x+1, y+1), (x+1, y-1), (x-1, y+1), (x-1, y-1).

(x-1, y+1)		(x+1, y-1)
	P (x,y)	
(x-1, y-1)		(x+1, y+1)

These points, together with the 4-neighbors, are called the 8-neighbors of p, denoted by $N_8(p)$.

(x-1, y+1)	(x, y-1)	(x+1, y-1)
(x-1, y)	P (x,y)	(x+1, y)
(x-1, y-1)	(x, y+1)	(x+1, y+1)

As before, some of the points in $N_D(p)$ and $N_8(p)$ fall outside the image if (x,y) is on the border of the image.

Image sensing and Acquisition:

Fundamental Parameters of an Imaging System

Before you acquire images, you must set up your imaging system. Five fundamental parameters comprise an imaging system: resolution, field of view, working distance, sensor size, and depth of field. Figure 5 illustrates these concepts.

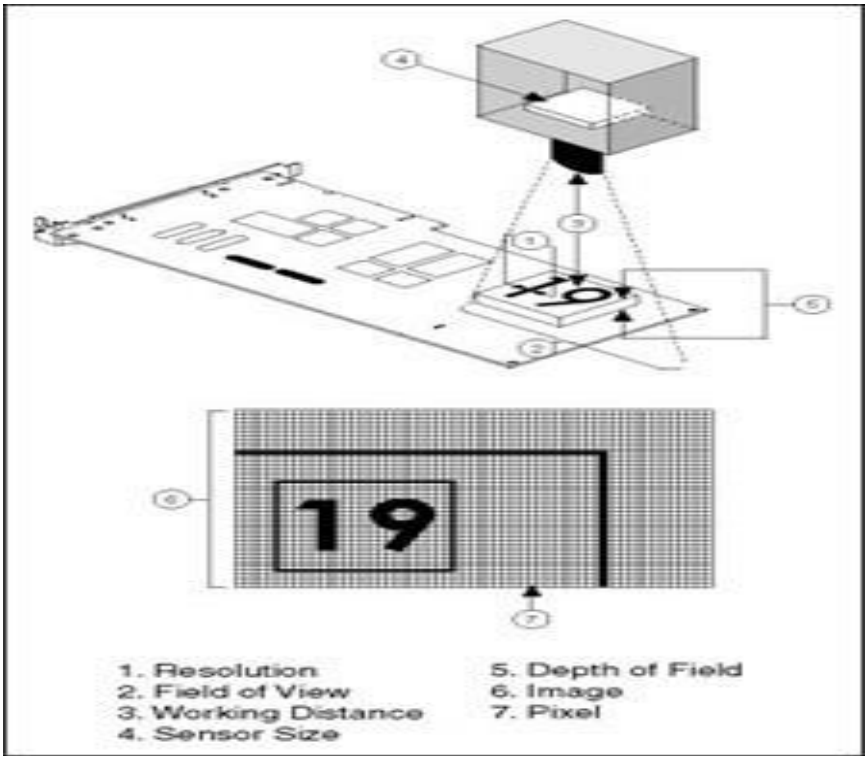


Figure 5a. Fundamental Parameters of an Imaging System

- Resolution: the smallest feature size on your object that the imaging system can distinguish
- Field of view: the area of inspection that the camera can acquire
- Working distance: the distance from the front of the camera lens to the object under inspection
- Sensor size: the size of a sensor's active area
- Depth of field: the maximum object depth that remains in focus

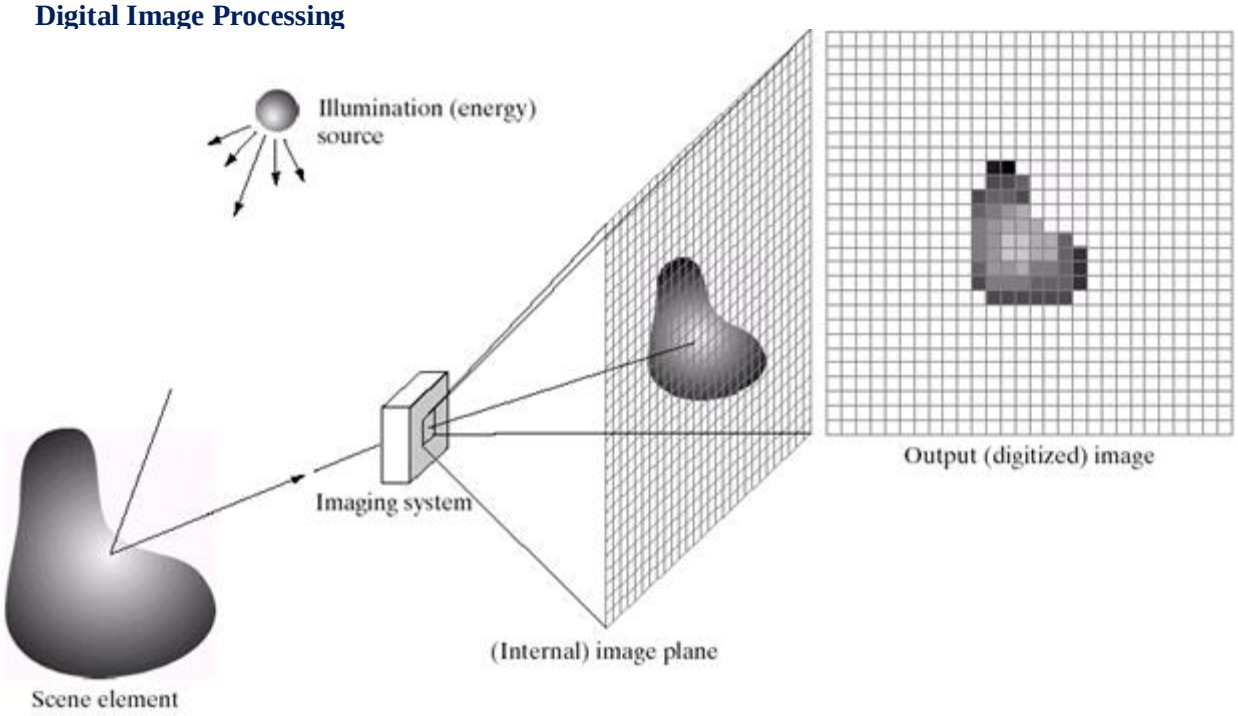
The way you set up your system depends on the type of analysis, processing, and inspection you need to do. Your imaging system should produce images with high enough quality to extract the information you need from the images. Five factors contribute to overall image quality: resolution, contrast, depth of field, perspective, and distortion.

The types of images in which we are interested are generated by the combination of an “illumination” source and the reflection or

absorption of energy from that source by the elements of the "scene" being imaged. We enclose *illumination* and *scene* in quotes to emphasize the fact that they are considerably more general than the familiar situation in which a visible light source illuminates a common everyday 3-D (three-dimensional)

scene.

For example, the illumination may originate from a source of electromagnetic energy such as radar, infrared, or X-ray energy. But, as noted earlier, it could originate from less traditional sources, such as ultrasound or even a computer-generated illumination pattern. Similarly, the scene elements could be familiar objects, but they can just as easily be molecules, buried rock formations, or a human brain. We could even image a source, such as acquiring images of the sun. Depending on the nature of the source, illumination energy is reflected from, or transmitted through, objects. An example in the first category is light reflected from a planar surface. An example in the second category is when X-rays pass through a patients body for the purpose of generating a diagnostic X-ray film. In some applications, the reflected or transmitted energy is focused onto a photo converter (e.g., a phosphor screen), which converts the energy into visible light. Electron microscopy and some applications of gamma imaging use this approach. The idea is simple: Incoming energy is transformed into a voltage by the combination of input electrical power and sensor material that is responsive to the particular type of energy being detected. The output voltage waveform is the response of the sensor(s), and a digital quantity is obtained from each sensor by digitizing its response. In this section, we look at the principal modalities for image sensing and generation.



a
b c d e

FIGURE An example of the digital image acquisition process. (a) Energy ("illumination") source. (b) An element of a scene. (c) Imaging system. (d) Projection of the scene onto the image plane. (e) Digitized image.

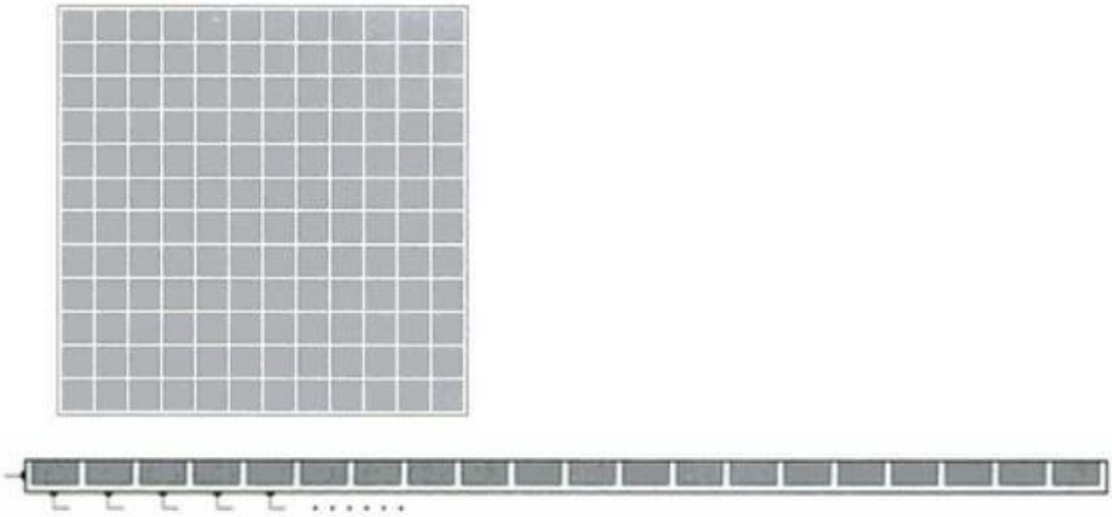


Fig5a: Line Sensor

Fig5b: Array- sensor

Lecture 3

Topic-SAMPLING AND QUANTIZATION

To create a digital image, we need to convert the continuous sensed data into digital from. This involves two processes – sampling and quantization. An image may be continuous with respect to the x and y coordinates and also in amplitude. To convert it into digital form we have to sample the function in both coordinates and in amplitudes.

Digitalizing the coordinate values is called **Sampling.** Digitalizing the amplitude values is called **Quantization**. There is a continuous the image along the line segment AB. To simple this function, we take equally spaced samples along line AB. The location of each samples is given by a vertical tick back (mark) in the bottom part. The samples are shown as block squares superimposed on function the set of these discrete locations gives the sampled function.

In order to form a digital, the gray level values must also be converted (quantized) into discrete quantities. So, we divide the gray level scale into eight discrete levels ranging from eight level values. The continuous gray levels are quantized simply by assigning one of the eight discrete gray levels to each sample. The assignment it made depending on the vertical proximity of a simple to a vertical tick mark. Starting at the top of the image and covering out this procedure line by line produces a two- dimensional digital image.

Image sampling and Quantization:

To create a digital image, we need to convert the continuous sensed data into digital form. This involves two processes.

1. Sampling and
2. Quantization

A continuous image, f(x, y), that we want to convert to digital form. An image may be continuous with respect to the x- and y- coordinates,

and also in amplitude. To convert it to digital form, we have to sample the function in both coordinates and in amplitude.
Digitizing the coordinate values is called **Sampling**. Digitizing the amplitude values is called **Quantization**.

Image Sampling and Quantization: The output of most sensors is a continuous voltage waveform whose amplitude and spatial behavior are related to the physical phenomenon being sensed. To create a digital image, we need to convert the continuous sensed data into digital form. This involves two processes: sampling and quantization.

The basic idea behind sampling and quantization is illustrated in Fig.6.1. Figure 6.1(a) shows a continuous image, f(x, y), that we want to convert to digital form. An image may be continuous with respect to the x- and y-coordinates, and also in amplitude. To convert it to digital form, we have to sample the function in both coordinates and in amplitude. Digitizing the coordinate values is called sampling. Digitizing the amplitude values is called quantization. The one-dimensional function shown in Fig.6.1 (b) is a plot of amplitude (gray level) values of the continuous image along the line segment AB in Fig. 6.1(a).

The random variations are due to image noise. To sample this function, we take equally spaced samples along line AB, as shown in Fig.6.1 (c).The location of each sample is given by a vertical tick mark in the bottom part of the figure. The samples are shown as small white squares superimposed on the function. The set of these discrete locations gives the sampled function. However, the values of the samples still span (vertically) a continuous range of gray-level values. In order to form a digital function, the gray-level values also must be converted (quantized) into discrete quantities. The right side of

Fig. 6.1 (c) shows the gray-level scale divided into eight discrete levels, ranging from black to white. The vertical tick marks indicate the specific value assigned to each of the eight gray levels. The continuous gray levels are quantized simply by assigning one of the eight discrete gray levels to each sample. The assignment is made depending on the vertical proximity of a sample to a vertical tick mark. The digital samples resulting from both sampling and quantization are shown in Fig.6.1 (d). Starting at the top of the image and carrying out this procedure line by line produces a two-dimensional digital image.

Sampling in the manner just described assumes that we have a continuous image in both coordinate directions as well as in amplitude. In practice, the method of sampling is determined by the sensor arrangement used to generate the image. When an image is generated by a single sensing element combined with mechanical motion, the output of the sensor is quantized in the manner described above. However, sampling is accomplished by selecting the number of individual mechanical increments at which we activate the sensor to collect data. Mechanical motion can be made very exact so, in principle; there is almost no limit as to how fine we can sample an image. However, practical limits are established by imperfections in the optics used to focus on the sensor an illumination spot that is inconsistent with the fine resolution achievable with mechanical displacements. When a sensing strip is used for image acquisition, the number of sensors in the strip establishes the sampling limitations in one image direction. Mechanical motion in the other direction can be controlled more accurately, but it makes little sense to try to achieve sampling density in one direction that exceeds the sampling limits established by the number of

sensors in the other. Quantization of the sensor outputs completes the process of generating a digital image.

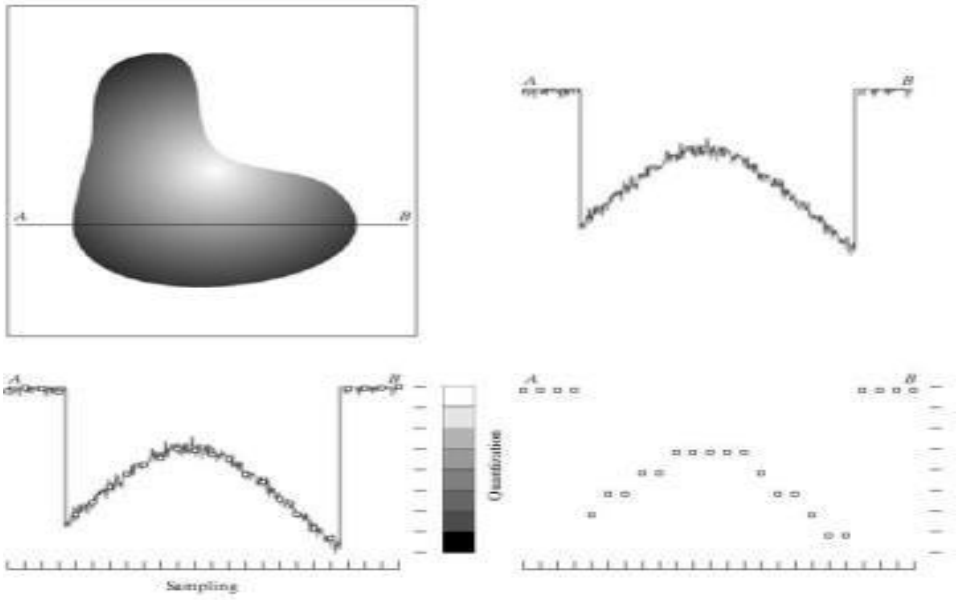
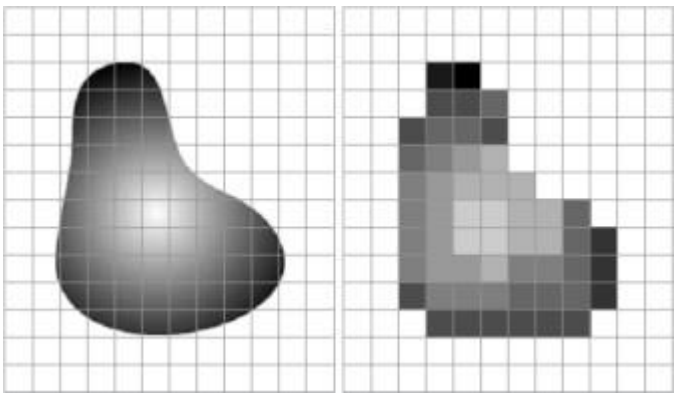


Fig6. Generating a digital image (a) Continuous image (b) A scan line from A to B in the continuous image, used to illustrate the concepts of sampling and quantization (c) Sampling and quantization. (d) Digital scan line

When a sensing array is used for image acquisition, there is no motion and the number of sensors in the array establishes the limits of sampling in both directions. Figure 6.2 illustrates this concept. Figure 6.2 (a) shows a continuous image projected onto the plane of an array sensor. Figure 6.2 (b) shows the image after sampling and quantization.

Clearly, the quality of a digital image is determined to a large degree by the



number ofsamples and discrete gray levels used in sampling and quantization.

Fig.6.2. (a) Continuous image projected onto a sensor array (b) Result of imagesampling and quantization.

Lecture 4

Topic-FUNDAMENTAL STEPS IN DIGITAL IMAGE PROCESSING:

There are two categories of the steps involved in the image processing –

1. Methods whose outputs are input are images.
2. Methods whose outputs are attributes extracted from those images.

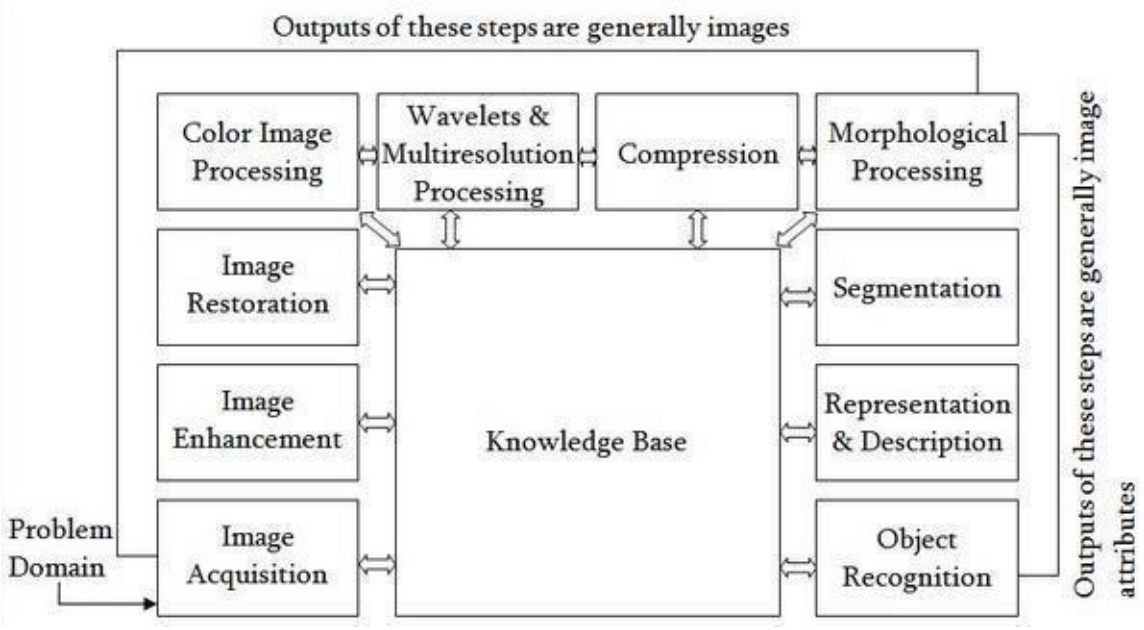
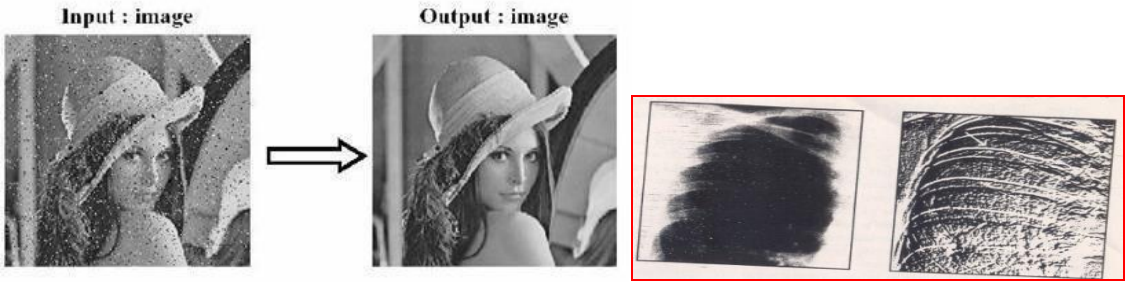


Fig7.: Fundamental Steps in Digital Image Processing

Image Acquisition: It could be as simple as being given an image that is already in digital form. Generally, the image acquisition stage involves processing such scaling.

Contrast Enhancement: It is among the simplest and most appealing areas of digital image processing. The idea behind this is to bring out details that are obscured or simply to highlight certain features of interest in image. Image enhancement is a very subjective area of image processing.



Fg,8 Sharpening

Image Processing: Techniques, Types, & Applications

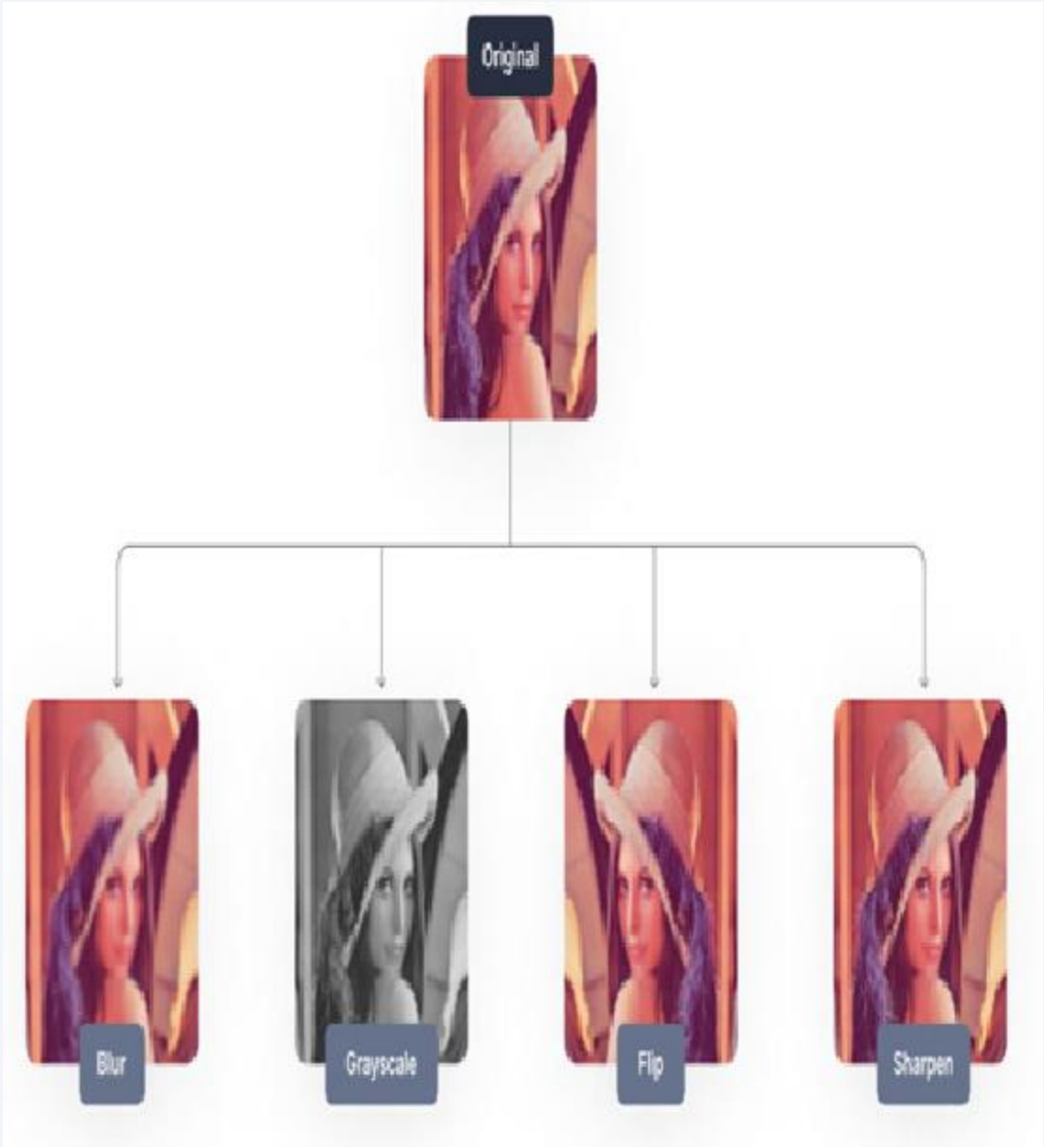


Fig.9 Basic operations

Digital Image processing is the class of methods that deal with manipulating digital images through the use of computer algorithms. It is an essential pre processing step in many applications, such as face recognition, object detection, and image compression.

Image processing is done to enhance an existing image or to sift out important information from it. This is important in several AI-based Computer Vision applications, where such pre processing can dramatically boost the performance of a model. Manipulating images, for example, adding or removing objects to images, is another application, especially in the entertainment industry.

Types of Images / How Machines "See" Images?

Digital images are interpreted as 2D or 3D matrices by a computer, where each value or pixel in the matrix represents the amplitude, known as the "intensity" of the pixel. Typically, we are used to dealing with 8-bit images, wherein the amplitude value ranges from 0 to 255.

1. Binary Image (Binarization)

Images that have only two unique values of pixel intensity- 0 (representing black) and 1 (representing white) are called binary images. Such images are generally used to highlight a discriminating portion of a colored image. For example, it is commonly used for image segmentation, as shown below.

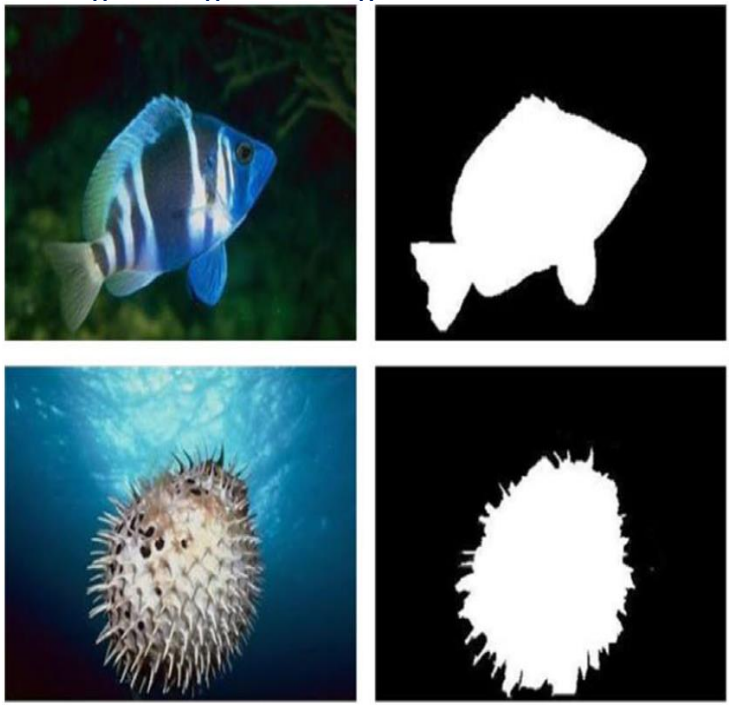


Fig.10 Binary image

2. Grayscale Image (Black & White Transformation)

Grayscale or 8-bit images are composed of 256 unique colors, where a pixel intensity of 0 represents the black color and pixel intensity of 255 represents the white color. All the other 254 values in between are the different shades of gray.

An example of an RGB image converted to its grayscale version is shown below. Notice that the shape of the histogram remains the same for the RGB and grayscale images.

3. RGB Color Image

The images we are used to in the modern world are RGB or colored images which are 16-bit matrices to computers. That is, 65,536 different colors are possible for each pixel. “RGB” represents the Red, Green, and Blue “channels” of an image.

Up until now, we had images with only one channel. That is, two coordinates could have defined the location of any value of a matrix. Now, three equal-sized matrices (called channels), each having values ranging from 0 to 255, are stacked on top of each other, and thus we require three unique coordinates to specify the value of a matrix element.

Thus, a pixel in an RGB image will be of color black when the pixel value is (0, 0, 0) and white when it is (255, 255, 255). Any combination of numbers in between gives rise to all the different colors existing in nature. For example, (255, 0, 0) is the color red (since only the red channel is activated for this pixel). Similarly, (0, 255, 0) is green and (0, 0, 255) is blue.

An example of an RGB image split into its channel components is shown below. Notice that the shapes of the histograms for each of the channels are different.

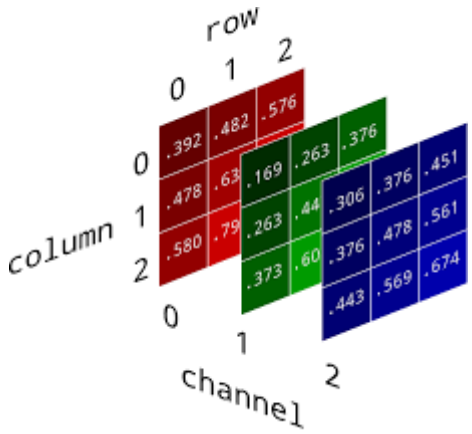


Fig. 10 Splitting of an image into its Red, Green and Blue channels

Phases of Image Processing

The fundamental steps in any typical Digital Image Processing pipeline are as follows:

1. Image Acquisition

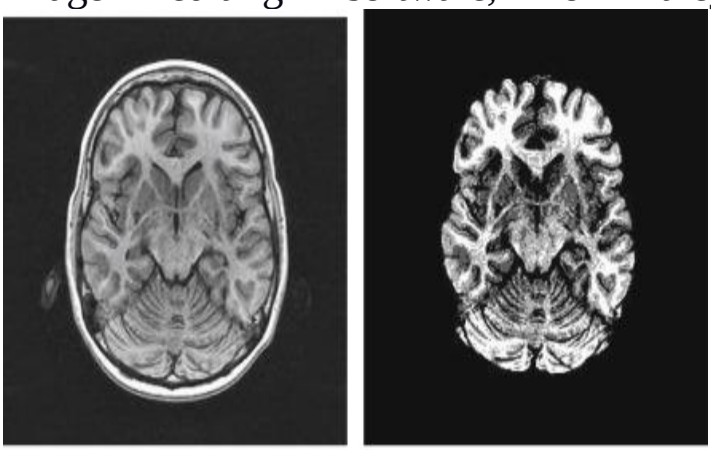
The image is captured by a camera and digitized (if the camera output is not digitized automatically) using an analogue-to-digital converter for further processing in a computer.

2. Image Enhancement **Task 1: Image Enhancement**

One of the most common image processing tasks is an image enhancement, or improving the quality of an image. It has crucial applications in Computer Vision tasks, Remote Sensing, and surveillance. One common approach is adjusting the image's contrast and brightness.

Contrast is the difference in brightness between the lightest and darkest areas of an image. By increasing the contrast, the overall brightness of an image can be increased, making it easier to see. Brightness is the overall lightness or darkness of an image. By increasing the brightness, an image can be made lighter, making it

easier to see. Both contrast and brightness can be adjusted automatically by most image editing software, or they can be adjusted manually.



(A) Brain MRImage (B) Afte Image enhancement and Skull Striping

However, adjusting the contrast and brightness of an image are elementary operations. Sometimes an image with perfect contrast and brightness, when upscaled, becomes blurry due to lower pixel per square inch (pixel density).

In this step, the acquired image is manipulated to meet the requirements of the specific task for which the image will be used. Such techniques are primarily aimed at highlighting the hidden or important details in an image, like contrast and brightness adjustment, etc. Image enhancement is highly subjective in nature.

3. Image Restoration

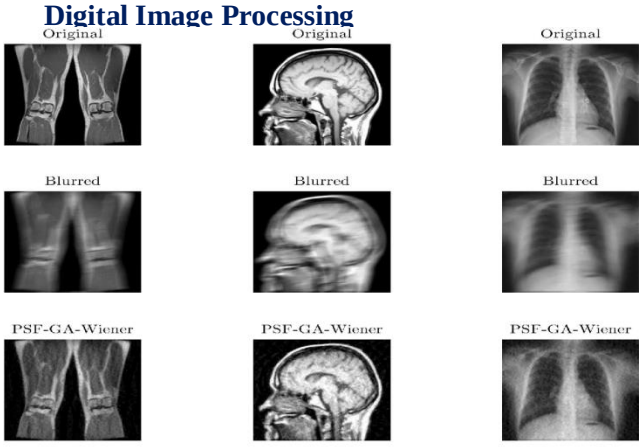
This step deals with improving the appearance of an image and is an objective operation since the degradation of an image can be attributed to a mathematical or probabilistic model. For example, removing noise or blur from images.

Task 2: Image Restoration

The quality of images could degrade for several reasons, especially photos from the era when cloud storage was not so commonplace. For example, images scanned from hard copies taken with old instant cameras often acquire scratches on them. Example of an Image Restoration operation

Image Restoration is particularly fascinating because advanced techniques in this area could potentially restore damaged historical documents. Powerful image restoration algorithms may be able to reveal large chunks of

missing information from torn documents



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Image Processing Techniques (Image inpainting)

Image processing can be used to improve the quality of an image, remove undesired objects from an image, or even create new images from scratch. For example, image processing can be used to remove the background from an image of a person, leaving only the subject in the foreground.

Image inpainting, for example, falls under this category, and it is the process of filling in the missing pixels in an image. This can be done by using a texture synthesis algorithm, which synthesizes new textures to fill in the missing pixels.

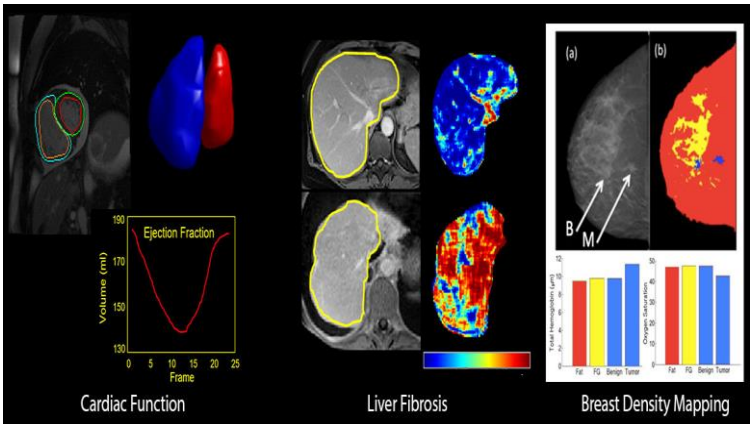


Example of an extreme image inpainting.

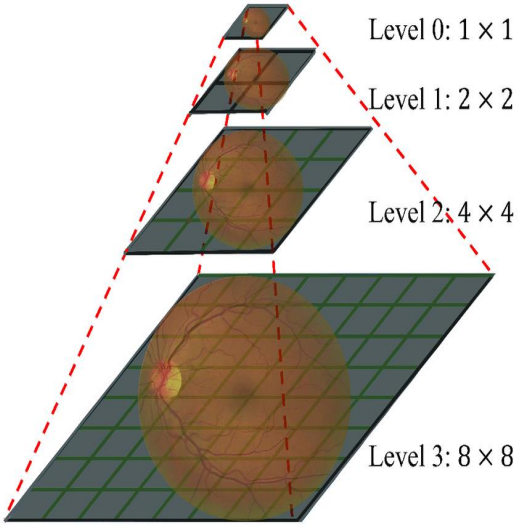
Inpainting is a conservation process where damaged, deteriorated, or missing parts of an artwork are filled in to present a complete image. This process is commonly used in image restoration

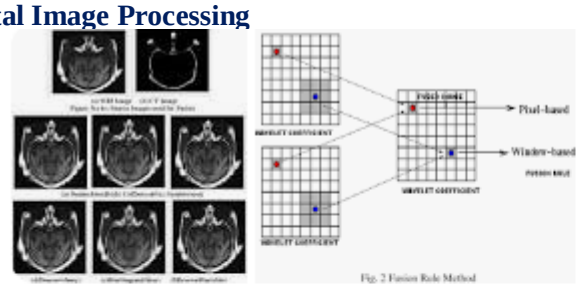
4. Color Image Processing

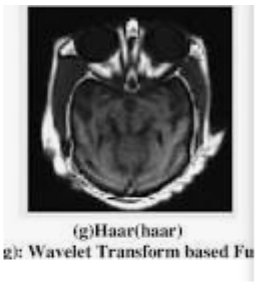
This step aims at handling the processing of colored images (16-bit RGB or RGBA images), for example, peforming color correction or color modeling in images.

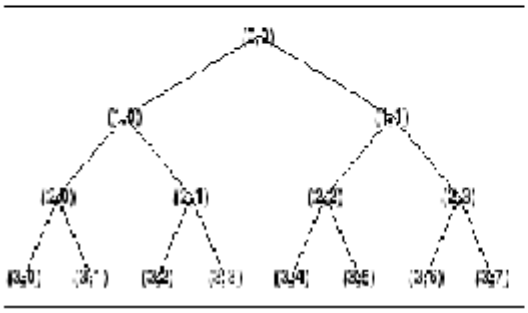


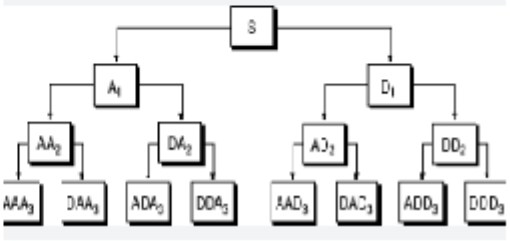
6. Multi-Resolution Processing









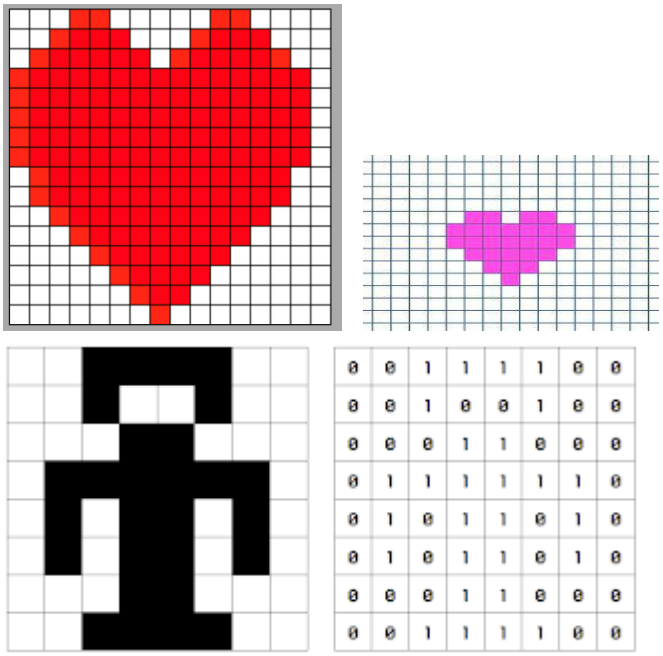


Wavelets are the building blocks for representing images in various degrees of resolution. Images subdivision successively into smaller regions for data compression and for pyramidal representation.

7. Image Compression

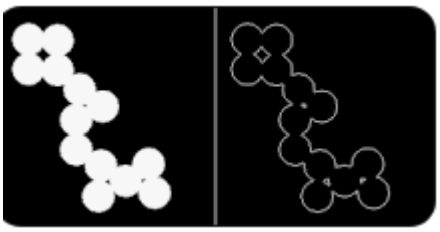
For transferring images to other devices or due to computational storage constraints, images need to be compressed and cannot be kept at their original size. This is also important in displaying images over the internet; for example, on Google, a small thumbnail of an image is a highly compressed version of the original. Only when you

click on the image is it shown in the original resolution. This process saves bandwidth on the servers.



7. Morphological Processing

Image components that are useful in the representation and description of shape need to be extracted for further processing or downstream tasks. Morphological Processing provides the tools (which are essentially mathematical operations) to accomplish this. For example, erosion and dilation operations are used to sharpen and blur the edges of objects in an image, respectively.



smoothing the boarder

8. Image Segmentation

This step involves partitioning an image into different key parts to simplify and/or change the representation of an image into something that is more meaningful and easier to analyze. Image segmentation allows for computers to put attention on the more important parts of the image, discarding the rest, which enables automated systems to have improved performance.



Task 3: Image Segmentation

Image segmentation is the process of partitioning an image into multiple segments or regions. Each segment represents a different object in the image, and image segmentation is often used as a preprocessing step for object detection.

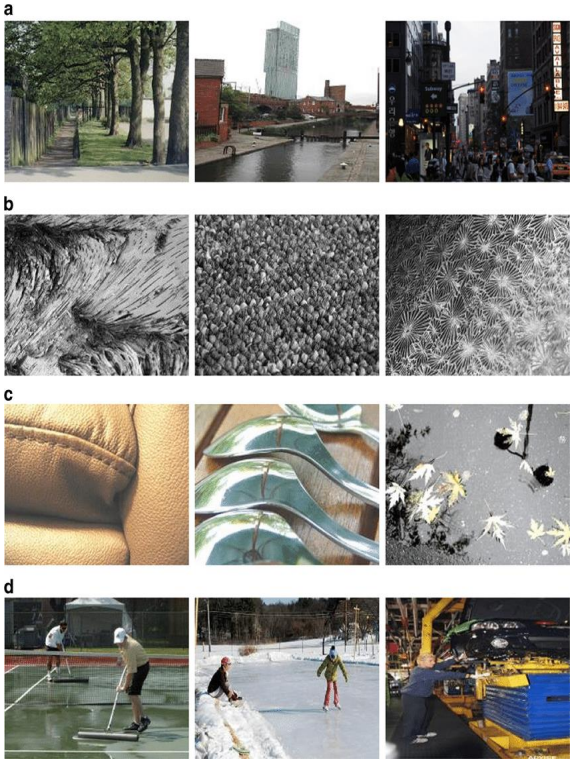
There are many different algorithms that can be used for image segmentation, but one of the most common approaches is to use thresholding. Binary thresholding, for example, is the process of converting an image into a binary image, where each pixel is either black or white. The threshold value is chosen such that all pixels with a brightness level below the threshold are turned black, and all pixels with a brightness level above the threshold are turned white. This results in the objects in the image being segmented, as they are now represented by distinct black and white regions.

Lecture 5

9. Representation and Description

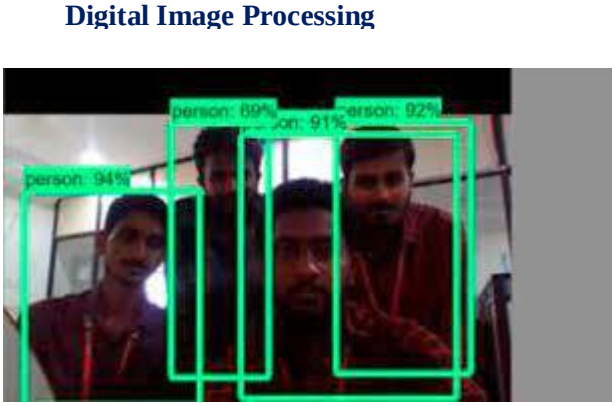
Image segmentation procedures are generally followed by this step, where the task for representation is to decide whether the segmented region should be depicted as a boundary or a complete region. Description deals with extracting attributes that result in some quantitative information of interest or are basic for differentiating one class of objects from another.

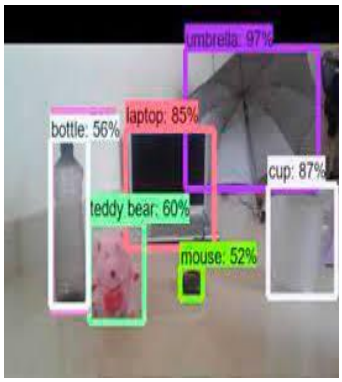




10. Object Detection and Recognition

After the objects are segmented from an image and the representation and description phases are complete, the automated system needs to assign a label to the object—to let the human users know what object has been detected, for example, "vehicle" or "person", etc.

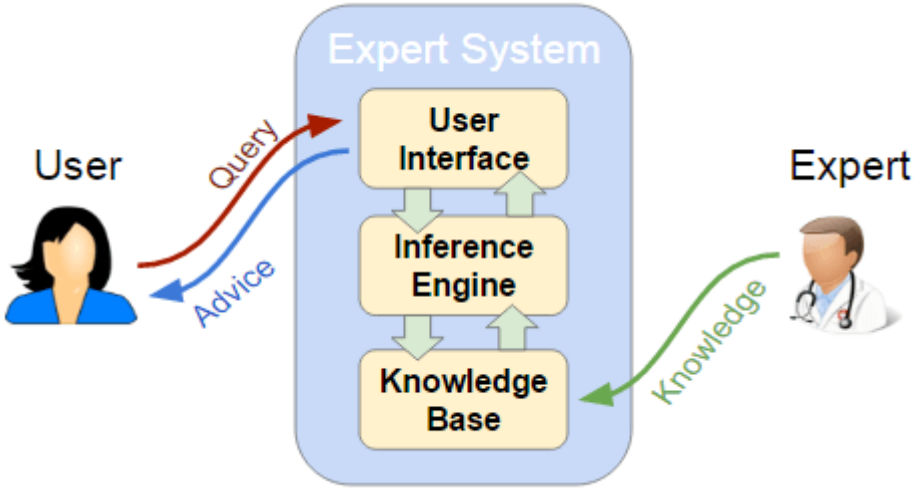






11. Knowledge Base

Knowledge may be as simple as the bounding box coordinates for an object of interest that has been found in the image, along with the object label assigned to it. Anything that will help in solving the problem for the specific task at hand can be encoded into the knowledge base.



Pixel: In a digital image, all the coordinates on 2-d function and the corresponding values are finite. Each value available in every location is considered as a pixel. In other words, a pixel is the smallest part of an image. So a digital image can be thought as 2-d array of pixels.

Gray Level: Each pixel has some intensity value which is called Gray level or Gray value. These values are usually represented in 8-bit int. So the range of values from 0 to 255. The values near to 0 indicates darker regions and the values near 255 represent brighter regions.

Types of Images: Binary, Grayscale, Color

Binary Image: A binary image has only two possible gray values or intensities 0 and 255, there are no intermediate values. Binary images are used as masks for indicating the pixels of interest in many image processing tasks. Below is the example of binary image.



Binary Image (only 0 and 255)

Grayscale Image: Grayscale image has range of values from 0 to 255 i.e, each pixel location can have any value

between 0 and 255. If you watch old films around the 1950s, you are watching grayscale images (films are nothing but videos which are collection of individual images in proper sequence). Here is the example below Grayscale images are monochrome images, Means they have only one color. Grayscale images do not contain any information about color. Each pixel determines available different grey levels.

A normal grayscale image contains 8 bits/pixel data, which has 256 different grey levels. In medical images and astronomy, 12 or 16 bits/pixel images are used.



Grayscale image (0–255 range)

Color Image: Both binary image and grayscale image are 2-dimensional arrays, where at every location, you have one value to represent the pixel. Remember to represent a color image, we need more than one value for each pixel. But how many values we need to represent a color? Typically you need 3 values for each pixel to represent any color. This came from the idea that any color can be formed by combining 3 basic colors Red, Blue and Green. Ex: you get yellow by mixing red and green, violet can be formed by combining red and blue etc., This is actually called RGB color space. There are many other ways to create color images which we will discuss in future discussions. Below is an example of a color image.

Colour images are three band monochrome images in which, each band contains a different color and the actual information is stored in the digital image. The color images contain gray level information in each spectral band.

The images are represented as red, green and blue (RGB images). And each color image has 24 bits/pixel means 8 bits for each of the three color band(RGB).

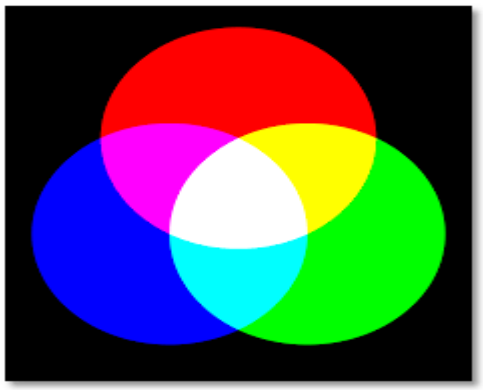
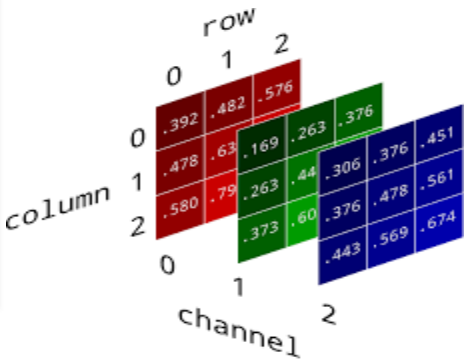
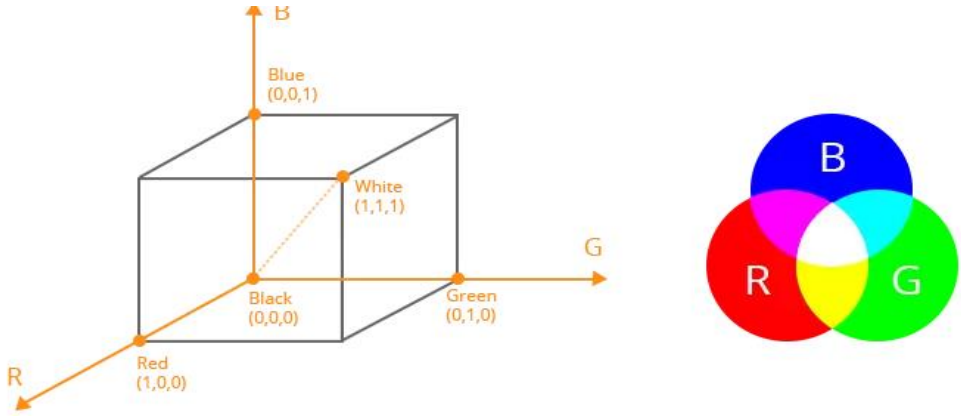
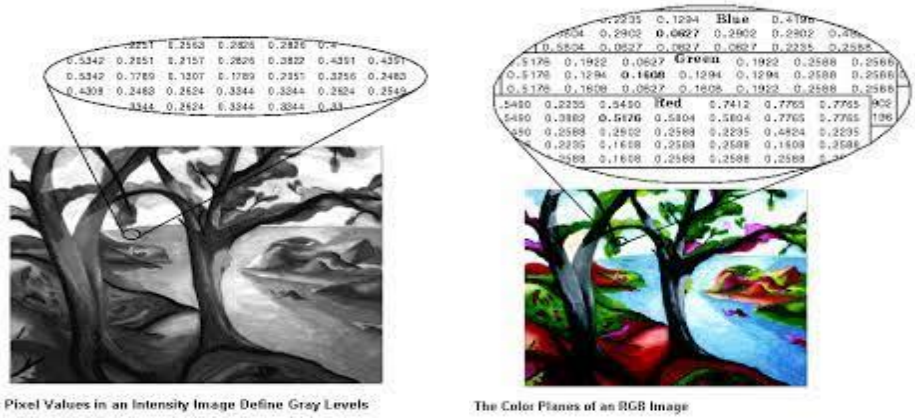


Fig. Colour image





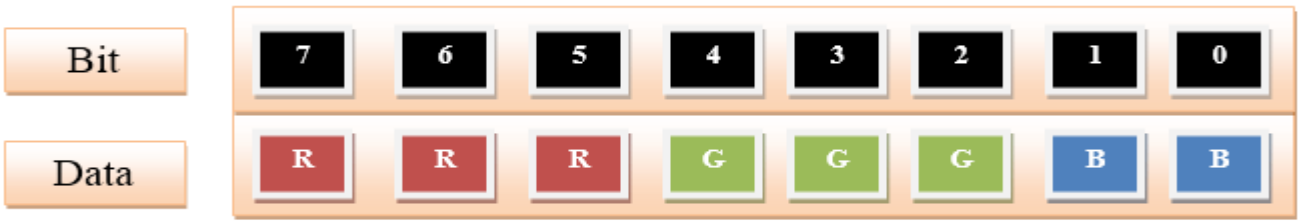


Pixel Values in an Intensity Image Define Gray Levels

The Color Planes of an RGB Image

8-bit color format

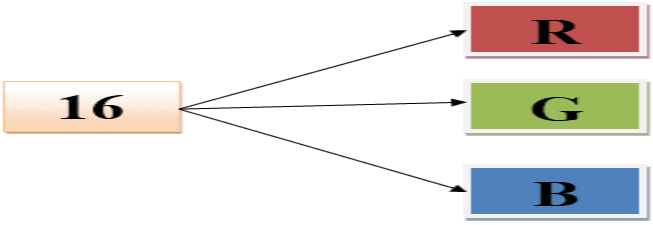
8-bit color is used for storing image information in a computer's memory or in a file of an image. In this format, each pixel represents one 8 bit byte. It has 0-255 range of colors, in which 0 is used for black, 255 for white and 127 for gray color. The 8-bit color format is also known as a grayscale image. Initially, it was used by the UNIX operating system.



16-bit color format

The 16-bit color format is also known as high color format. It has 65,536 different color shades. It is used in the system developed by Microsoft. The 16-bit color format is further divided into three formats which are Red, Green, and Blue also known as RGB format.

In RGB format, there are 5 bits for R, 6 bits for G, and 5 bits for B. One additional bit is added in green because in all the 3 colors green color is soothing to eyes.



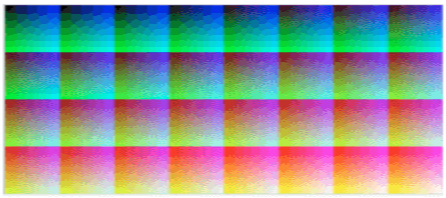


Fig: RGB 16 bits palette

24-bit color format

The 24-bit color format is also known as the true color format. The 24-bit color format is also distributed in Red, Green, and Blue. As 24 can be equally divided on 8, so it is distributed equally between 3 different colors like 8 bits for R, 8 bits for G and 8 bits for B.

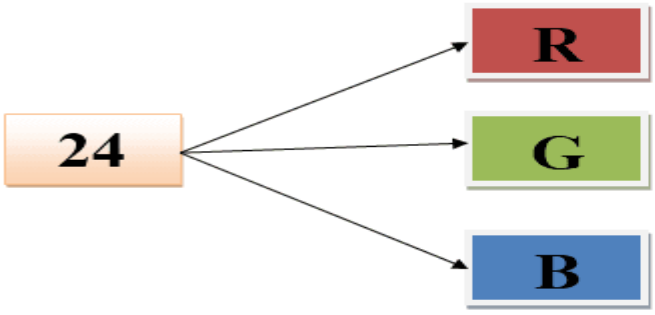




Fig: 16,777,216 colors

Video: A video is just a collection of images in a proper sequence.

Image formats-

1. **JPEG (or JPG) - Joint Photographic Experts Group**
2. **PNG - Portable Network Graphics**
3. **GIF - Graphics Interchange Format**
4. **TIFF - Tagged Image File**
5. **PSD - Photoshop Document**
6. **PDF - Portable Document Format**
7. **EPS - Encapsulated Postscript**
8. **AI - Adobe Illustrator Document**
9. **INDD - Adobe Indesign Document**
10. **RAW - Raw Image Formats**